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Detection of Alcohol-Impaired Driving using Eye-Tracking and Vehicle Data via Machine Learning

--Manuscript Draft--

Full Title:	Detection of Alcohol-Impaired Driving using Eye-Tracking and Vehicle Data via Machine Learning
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DETECTION OF ALCOHOL-IMPAIRED DRIVING USING EYE-TRACKING AND VEHICLE DATA VIA MACHINE LEARNING

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ABSTRACT

Detection of alcohol impairment using driver monitoring systems (DMS) is an active area of research, with technologies under development that rely upon sensors, cameras, and other detection mechanisms. We analyze the capacity of ocular parameters from a camera-based DMS and vehicle-input data to classify data from sober and alcohol-intoxicated drivers collected using a validated driving simulator. Our approach split the time-series produced by the DMS and simulator into intervals of various durations ranging from 15 to 90 seconds and trained machine learning methods using eye-only, vehicle-only, and combined eye and vehicle inputs to classify these intervals. Models were evaluated using the mean area under the receiver operating characteristic curve (AUC) across ten randomly assigned training-validation splits. We found feature learning using Random Convolutional Kernel Transform (ROCKET) models trained on 90-seconds of combined vehicle and eye data resulted in the best classification performance (AUC=0.995, range=0.906-0.986). Overall, classification performance increased with longer duration intervals, and models trained using both eye and vehicle inputs exhibited better performance than those using either eye-only or vehicle-only data. when targeting exactly zero false positive classifications in our validation data we found that in excess of 40% of true positives can be correctly identified by several of our models. Using feature importance methods, we found features representing higher-order moments of the pupil size distribution (skew and kurtosis) to be most influential. These results demonstrate the potential of camera-based DMS and vehicle-data to be used in tandem to accurately identify instances of alcohol impairment in select driving environments.

Keywords: Impaired Driving, Alcohol, Driver Monitoring, Impaired Driver Detection

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INTRODUCTION

Alcohol-impaired drivers were responsible for approximately 30 percent of all vehicle fatalities in the United States in 2023 (*1*), making the identification of alcohol-impaired driving a priority. However, the preventative potential of impairment assessments including breathalyzers, blood tests, and field sobriety tests is limited, as these assessments require a driver to have been identified and stopped by law enforcement. In 2023, the National Highway Traffic Safety Administration (NHTSA) published an Advanced Notice of Proposed Rule Making aiming to set standards for technology to prevent impaired driving in 2023 (*2*), prompting increased research attention to alcohol detection using driver monitoring systems (DMS).

The development of alcohol detection systems is built upon a substantial body of research demonstrating that alcohol and other drugs cause detectable changes in ocular behavior that can be measured using gaze tracking devices (*3*). Similarly, numerous experimental studies have shown have shown that drug impairment influences driving performance measures (*4*). Taken together, these results suggest the potential for DMS relying upon ocular and vehicle data to accurately identify instances of alcohol-impaired driving.

A handful of previous studies have developed machine learning systems to detect drunk driving using DMS and vehicle-data, some of which utilize driving simulators and some of which using real vehicles on test tracks Koch et al. (5), Deuber et al. (6). Other examples include Çetinkaya and Acarman (7), who used tree-based feature selection and gradient boosting to classify drivers as impaired using eye glances and vehicle sensors, McDonald et al. (8), who applied a random forest algorithm to wheel-angle data to detect instances of drowsiness, and Dong et al. (9), who used non-intrusive image sequences of a driver's face to detect fatigued and distracting driving using a random forest algorithm and neural network models.

One limitation of many of these studies, and an important practical consideration when applying machine learning models aimed at detecting alcohol-impaired driving, is the challenge in moving from classification of alcohol-impaired driving using fixed duration samples to continuous monitoring. Our research addresses this limitation by analyzing the performance of modern machine learning methods over several different measurement intervals ranging from 15-seconds to 90-seconds. We use data from an simulated driving study where participants were dosed with alcohol and drove simulated routes with an automotive-grade DMS recording eye and face data such as head pose, gaze direction and eye closure. The simulator platform recorded various driver inputs and vehicle states, facilitating an investigation of the relative contributions of eye and vehicle data towards accurate classifications of alcohol impairment using machine learning. Our work contributes to the growing literature on alcohol detection using DMS through its use of modern machine learning methods and advanced eye-tracking systems to assess classification performance over varying time intervals.

METHODS

Data Collection

Participants recruited through a large email registry and word of mouth took part in a driving simulator study consisting of a baseline drive, alcohol dosing, and four post-dose drives. After screening for recent drug use, pregnancy, physical or psychiatric conditions, and motion sickness,

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N=36 individuals completed the study. The Sahlgrenska formula was used to calculate the alcohol dose needed to bring each individual's breath alcohol concentration (BrAC) to a target peak of 0.115. The dose was mixed with orange juice and split evenly among three drinks, which were consumed over consecutive 10-minute periods such that the entire dose was consumed in 30-minutes. BrAC concentrations were measured every 5 to 10-minutes during the dosing period using a calibrated Alco-Sensor FST breath alcohol testing instrument to ensure the target BrAC was reached. The four post-dose drives occured at BrAC concentrations of 0.1, 0.085, 0.07, and 0.055 with an allowed margin of error of ± 0.01. BrAC concentrations were continually monitored and if a subject's levels were declining faster than expected a small secondary dose of alcohol was administered to bring them back into the target BrAC range for the next drive. Subjects also completed a short questionnaire before and after each drive that included their self-reported readiness to drive (yes/no) and Karolinska Sleepiness Scale (KSS) score (*10*), a validated fatigue scale ranging from 0-9.

Simulated drives were performed in a miniSim™ (*11*), a validated PC-based driving simulator featuring a a quarter cab with three 48-inch, high-definition (1080p) LCD providing a comprehensive forward field of vision. Drives had a duration of approximately 25-minutes and featured a mixture of straight and curved sections of rural roadways in a nighttime environment with variable amounts of traffic. The scenario order and route variation of each drive were randomly assigned to minimize learning effects. An automotive-grade DMS produced by Seeing Machines Ltd (*12*) was integrated into the driving simulator to simultaneously collect data on eye behavior during each drive. The DMS uses near-infrared lighting and an automotive grade camera focused on the driver's face to capture ocular parameters that can be used to estimate driver state. The system processes camera images to extract eye and face data such as head pose, gaze direction, and eye closure. Each participant's facial shape was matched to a preset shape to facilitate accurate head and eye-tracking by the DMS. Data from the DMS and miniSim™ were aligned to produce a 60 Hz time-series of the participant's vehicle states, driver inputs, and their head and eye positions throughout each drive.

Data Preparation

Our analyses use data from a roughly 8-minute segment straight rural road encountered near the end of each drive. We excluded 27 of 180 drives due to incompatibilities or missing data. 10 exclusions were due to corrupted eye or vehicle data and 17 did not contain at least 90-seconds of driving in the rural straight segment. The remaining 153 drives were segmented into intervals using a sliding window approach that allowed 50% overlap between consecutive intervals. This was done for interval lengths of 15, 30, 45, 60, 75, and 90 seconds. Intervals were then assigned to be used for either training or validation. Due to the overlapping nature of intervals within the same drive, all intervals within a drive were randomly assigned to training or validation in order to prevent data leakage. This training-validation process was repeated ten times using different random assignments to ensure the performance metrics we report are robust to variation introduced by splitting the data into training and validation sets.

When developing machine models to classify intervals as coming from an alcohol-dosed or sober sober drive we adopt two general approaches: feature engineering and feature learning. For the first approach, we created features from eye and vehicle inputs based upon relationships reported

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in the literature on alcohol and ocular behavior while drive (*13–18*) as well as documented relationships between vehicle measures and drug impairment (*19, 20*). This approach is similar to the methods described in (*5, 6*), whose work also focused on the development of machine learning models for the detection of alcohol-impaired driving. For each selected variable we calculate the mean, standard deviation, 5th and 95th percentile, range between the 5th and 95th percentile, skewness, and kurtosis over the time interval of the sample. Several additional variables were calculated for inputs that could not naturally be aggregated across time-steps, including count related features such as the number of blinks, saccades, and fixations. We use the 'saccades' package in 'R' (*21*) to calculate these features for each interval. The package uses a velocity-based detection algorithm with reliable performance on eye-tracking data with frequencies below 100Hz (*22*). A complete list of all engineered features can be found in Table 1. In total, 116 eye-related and 40 vehicle-related features were engineered from the raw data.

Machine Learning Models

The purpose of feature engineering is to reduce each interval to a single data-point without a time dimension so that it can be used an input into a conventional machine learning model. For this we consider random forest (*23*) and gradient boosting as implemented in the popular 'XGBoost' library (*23*). Both random forest and gradient boosting are ensemble models that are built from a large number of decision trees. This structure allows the models to learn complex non-linear relationships and interactions from the input features. Additionally, the influence of individual features within these models can be quantified by importance metrics that are based upon the purity gains attributed to splits involving particular features. Hyperparamters for these models were chosen separately for each training-validation split using a cross-validated grid search performed on the training data. For random forest models we optimized the number of trees, number of features considered at each split, and maximum tree depth. For 'XGBoost' models we optimized the learning rate, number of trees, maximum tree depth, maximum number of features considered at each split, and degree of L1 regularization.

Our second modeling approach, feature learning, used the Random Convolutional Kernel Transform (ROCKET) methodology first proposed by (*24*). ROCKET uses a large number of convolutional kernels, 10000 in our application, with randomly generated weights to produce a set of feature maps that capture the temporal patterns in each input represented by the randomly generated weights. Each feature map is then summarized by its percentage of positive values (PPV) and absolute maximum value (AM). The resulting collection of PPVs and AMs are then used as inputs in a final classification model, regularized logistic regression in our application. In contrast with our feature engineering approach, ROCKET learns predictive patterns on its own from the time-series inputs.

Model Evaluation

We evaluated all of our machine learning models using receiver operating characteristic (ROC) curves. An ROC curve displays the true positive rate (TPR) and false positive rate (FPR) for every different classification threshold, thereby providing a holistic view of model performance that isn't influenced by an arbitrary threshold for classifying an interval as alcohol-impaired. We used the

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area under the ROC curve (AUC) to summarize the overall performance of each model we considered. Unless otherwise stated, we report AUC values that are averaged across our ten randomly chosen training-validation splits.

While AUC provides a holistic assessment of model performance across a range of classification thresholds, a model must use a fixed threshold in order to be used in practice. To address this consideration, we also report the true positive rate for the smallest classification threshold that produced exactly zero false positive classifications in a given training-validation split.

Separate models were trained and evaluated for each combination of modeling approach (random forest, XGBoost, or ROCKET), type of input features made available to the model (eye-only, vehicle-only, or both vehicle and eye), and the duration of intervals used for training and evaluation (15, 30, 45, 60, 75, and 90 seconds). We also report feature importance metrics for the top-performing models in our featuring engineering approach.

RESULTS

Demographic characteristics of the N=36 individuals who completed the study are displayed in Table 2. The sample contained an equal number of males and females, individuals from a variety of racial and ethnic backgrounds, and individuals with various different skin tones, a relevant consideration for eye-tracking technology.

Overall, we found that ROCKET models using both eye and vehicle inputs measured over 90-second time intervals produced the best classification performance, with a mean AUC of 0.955 across validation sets (SD = 0.027, range = 0.906-0.986). Using only eye-related inputs, ROCKET achieved a mean AUC of 0.916 (SD = 0.039, range = 0.842-0.977), and using only vehicle-related inputs ROCKET achieved a mean AUC of 0.880 (SD = 0.034, range = 0.818-0.925). Results for all models on 90-second intervals are provided in Table 3.

Model Comparisons

The mean AUC across validation sets by interval length and modeling approach is displayed in Figure 1. In general, models perform best when trained on longer duration intervals, with all approaches exhibiting an upward trend in performance as interval duration increases. Each model approach achieved its best performance when using both eye and vehicle data, and ROCKET models tended to exhibit the best performances overall. For the two feature engineering approaches (random forest and XGBoost) models using only vehicle-inputs outperform those using only eye-related inputs. However, the opposite relationship is observed for ROCKET models, with eye-only models outperforming vehicle-only models. ROCKET models trained using only eye-related features also outperformed the feature engineering approaches that used both eye and vehicle features.

ROC curves for models trained on 30, 60, and 90 second intervals are shown in Figure 2. These curves lend further support to the superior performance of ROCKET models when both vehicle and eye, or eye-only features are used as inputs. When only vehicle inputs are used ROCKET shows slightly better performance when using 90-second intervals, but performs similarly to the

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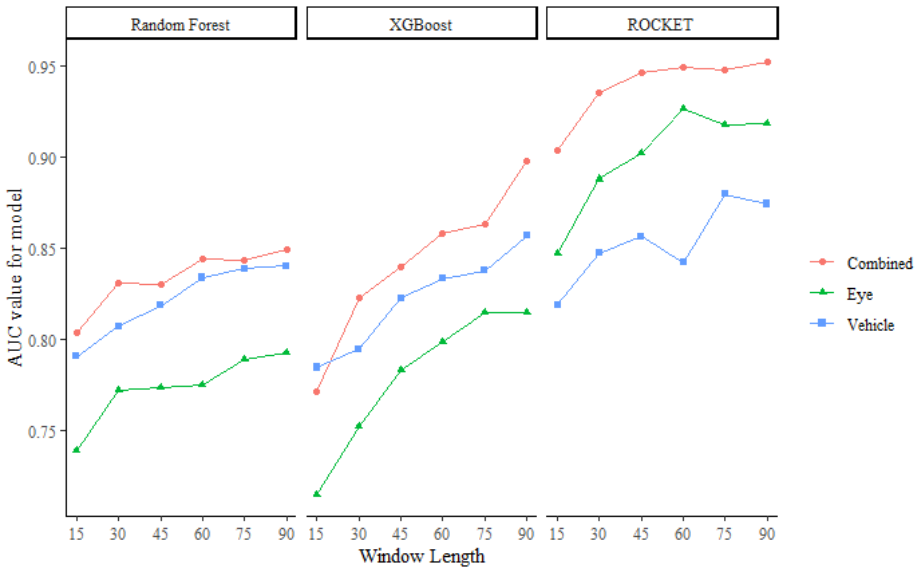


FIGURE 1 Average AUC values across models, the predictors used for each model, and window lengths.

feature engineering approaches when shorter duration intervals are used.

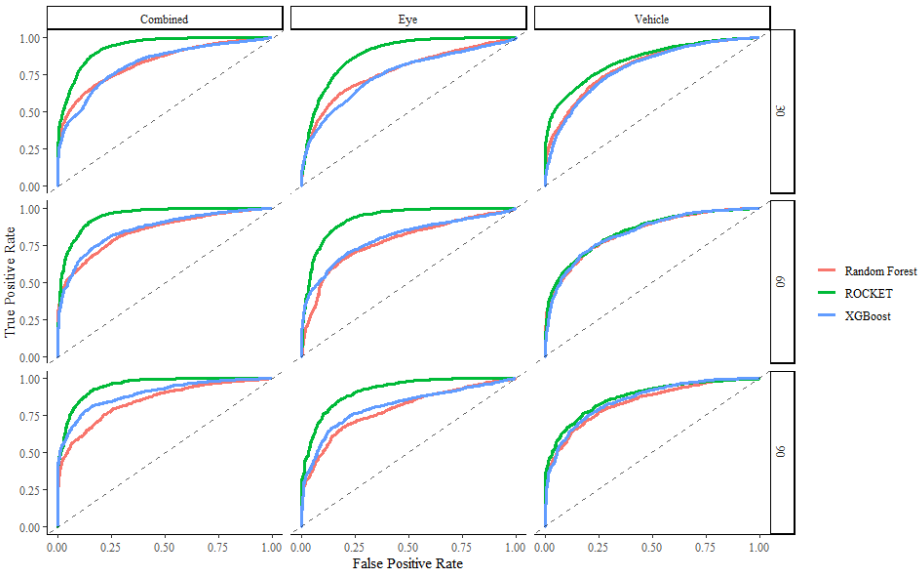


FIGURE 2 ROC Curves by predictor type and model across selected window lengths(30s,60s, and 90s)

The maximum TPR achieved by model trained on 90-second intervals using both eye and vehicles inputs at the decision threshold yielding exactly zero false positives across training-validation splits is displayed in Figure 3. Despite the dominance of ROCKET models displayed in previous figures, XGBoost tends to produce the highest median TPR for each interval duration, though it

also exhibits the greatest amount of variability across splits. Altogether, these results suggest that true positive rates in excess of 40% with a minimal number of false positives can be achieved when using vehicle and eye inputs measured over sufficiently long time intervals.

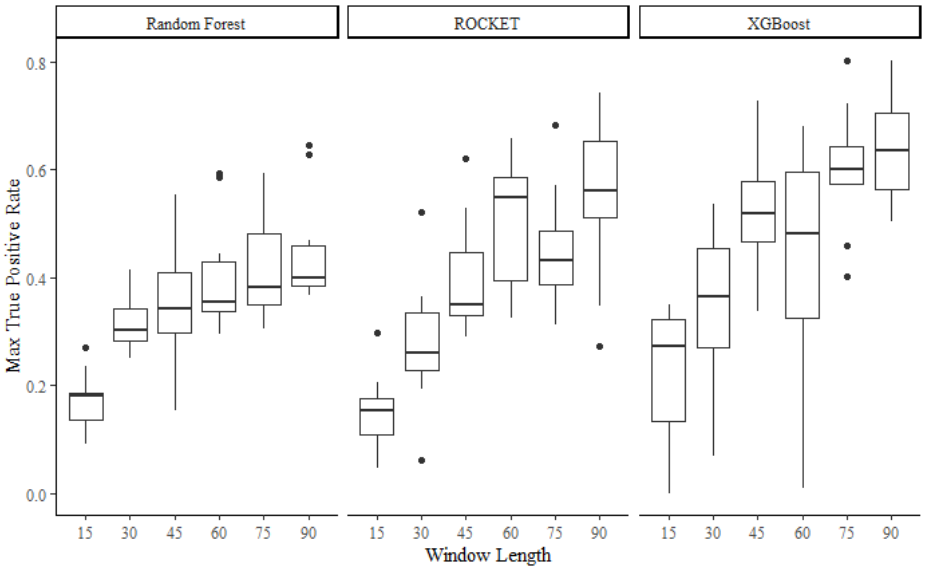


FIGURE 3 Max TPR (True Positive Rate) before first False Positive across combined predictor models.

Feature Importance

Despite being outperformed by ROCKET models, an advantage of our feature engineering models is the capacity to understand which inputs are most influential in the model's classification decisions. Figure 4 displays the total importance of eye and vehicle derived features for XGBoost models by interval length. Because these models used nearly three times as many eye-related features (116 versus 40), we include raw, non-standardized results as well as results that are standardized by the number of available features of each type. The relative importance of features derived from eye and vehicle data does not vary by interval duration, and eye-related features are more influential in aggregate, but less influential on a per-feature basis.

The four most influential features for XGBoost models are summarized in Table 4. Higher moment features (skewness and kurtosis) derived from pupil size exhibit the greatest importance in models trained using eye-only and combined vehicle and eye data. Head position features also tend to make notable contributions across models. The most influential vehicle-based features tend to capture variability in steering and lane keeping, with the standard deviation of steering wheel angle rate of change consistently appearing as the top contributor.

Across interval lengths, eye-related features occupy 10 of 24 spots among the top four most influential features, which is consistent with the greater importance of vehicle-based data on a per-feature basis. However, pupil skewness is consistently the most influential feature across all interval dura-

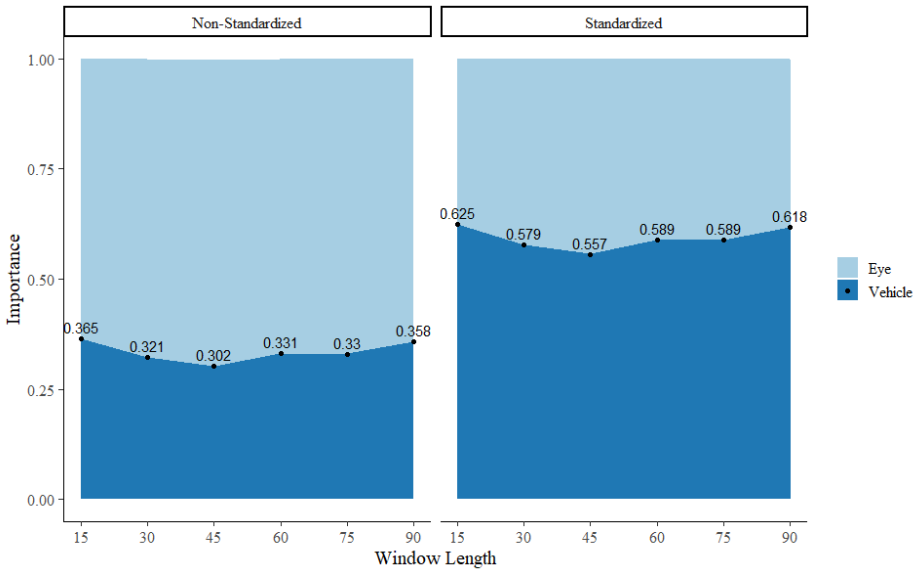


FIGURE 4 Aggregate contributions of eye and vehicle derived features to XGBoost models trained on 90-second samples. Non-standardized contributions reflect the raw total contribution of all features, while standardized contributions re-scale the contributions proportional to the number of eye and vehicle features.

tions, and its importance remains consistently high across all training-validation splits.

DISCUSSION

Overall, we found the optimal use of machine learning methods in the automated detection of alcohol impairment depends upon the underlying classification goals. When very stringent control over the false positive rate is desired, we found that handcrafted features derived from both eye and vehicle data measured over modest time intervals ($\geq$ 45-seconds) produced true positive rates exceeding 40%. However, ROCKET, a feature learning model, trained using a combination of eye and vehicle inputs yielded the best mean AUC values by a wide margin across all interval durations. Averaged over 10 randomly assigned training-validation splits, ROCKET achieves an AUC of 0.955 for 90-second intervals, a performance that exceeds those reported in similar studies (*5, 6*). The ability of ROCKET to outperform our feature engineering approaches, particularly for eye-related features, suggests there are predictive temporal patterns that likely are not being captured by the features we created. Future research should further investigate these patterns and their capacity to be encoded into a feature engineering approach.

We found eye and vehicle information to be complimentary, with various features from each source being among the most influential predictors in feature engineering models, and models that use features from both eye and vehicle data outperforming models trained on only eye or only vehicle data. This suggests that DMS should consider a diverse range of inputs when seeking maximize detection ability. Future work should consider combinations of features from additional data streams such as heart rate or other physiological features.

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Limitations

Our analyses rely upon data from a single driving environment consisting of approximately 8-minutes on a straight rural roadway. While the detection of alcohol impairment on rural roads is important, further research is needed to assess the generalizability of our findings to more complex driving environments. Furthermore, our data from a driving simulator setting where factors such as weather, lighting, glare, and the presence of obstructive objects are controlled. Future work should investigate the robustness of alcohol detection methods to variation in these factors.

The driving simulator used in this research did not incorporate features common to on-road vehicles including adaptive driver-assistance systems (ADAS), lane keeping assistance, and cruise control. The prevalence of these features highlights the importance of accurate models that can rely exclusively upon eye-related features. However, it remains uncertain how certain eye behaviors might differ when an individual is driving a vehicle with the assistance of ADAS.

CONCLUSION

Our results demonstrate the potential for DMS using eye and vehicle data to accurately identify instances of impaired driving. We find that measuring eye and vehicle features over longer time intervals generally yields better performance, though the AUC achieved by ROCKET, our top performing model, plateaus once roughly 45-60 seconds of data is used. We found eye-related features to be most influential in our models, with higher moment descriptors of the distribution of pupil size being most impactful. Nevertheless, vehicle-based features made meaningful contributions towards accurate identification.

DECLARATION OF CONFLICTING INTERESTS

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: Ryan Miller, Timothy Brown; data collection: Timothy Brown; data reduction and preparation: Andy Chestovich, Anshul Satav; analysis and interpretation of results: Ryan Miller, Andy Chestovich, Anshul Satav, Timothy Brown; draft manuscript preparation: Ryan Miller, Andy Chestovich, An-

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shul Satav, Timothy Brown; All authors reviewed the results and approved the final version of the manuscript.

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Feature group	Feature subgroup	Measures	Variable Name	Description
Eye	Eyelid behavior	Proportions and Counts	Left PERCLOS	Proportion of time left eyelid closed $> 80\%$
			Right PERCLOS	Proportion of time right eyelid closed $> 80\%$
			Blinks	Number of blinks detected
		Full summary statistics	Blink Duration	Duration of blinks
	Pupil characteristics	Full summary statistics	Left Pupil Diameter	Left pupil diameter (mm)
			Right Pupil Diameter	Right pupil diameter (mm)
	Head position	Full summary statistics	Head Position X	Head position side-to-side
			Head Position Y	Head position up-down
			Head Position Z	Head position front-back
	Gaze direction	Proportions	PERLOOK	Proportion of time eyes off road
		Full summary statistics	Left Gaze X	X-coordinate left eye gaze vector
			Left Gaze Y	Y-coordinate left eye gaze vector
			Left Gaze Z	Z-coordinate left eye gaze vector
			Right Gaze X	X-coordinate right eye gaze vector
			Right Gaze Y	Y-coordinate right eye gaze vector
			Right Gaze Z	Z-coordinate right eye gaze vector
			Unified Gaze Pitch	Both eyes gaze pitch angle (deg)
			Unified Gaze Yaw	Both eyes yaw angle (deg)
	Saccade & fixation	Mean	Fixations	Number of fixations
			Fixation Duration	Duration of fixations
		Mean & Std dev	Horizontal Dispersion	Horizontal spread of gaze
			Vertical Dispersion	Vertical spread of gaze
			Peak Velocity Horizontal	Max horizontal saccade velocity
			Peak Velocity Vertical	Max vertical saccade velocity
Vehicle	Dynamics & control	Full summary statistics	Vehicle Speed	Speed of vehicle
			Brake Force	Force on brake pedal
			Steering Angle	Steering wheel angle
			Steering Angle Rate	Rate of steering wheel angle change
			Lane Deviation	Deviation from lane center
			Acceleration Pedal	Accelerator pedal position

TABLE 1 Eye and Vehicle Variable Features with Signal Types and Aggregation Methods

Note: Full summary statistics include: Mean, Standard deviation, 5th & 95th percentiles, Skewness, Kurtosis, and Middle 90th percentile.

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Variable	Distribution
Age	21–65 years (mean = 30.9, *SD* = 10.8)
Sex	Male = 50%, Female = 50%
Hispanic	Yes = 4 (11.1%) No = 32 (88.9%)
Race	Asian/Asian American = 6 (16.7%) Black/African American = 3 (8.3%) White = 23 (63.9%) More than one race = 2 (5.6%) Prefer not to answer = 2 (5.6%)
Skin tone—Fitzpatrick scale (*25*)	Type 1 = 0 (0%) Type 2 = 25 (69.4%) Type 3 = 8 (22.2%) Type 4 = 4 (11.1%) Type 5 = 3 (8.3%) Type 6 = 1 (2.8%)

TABLE 2 Sample demographic characteristics.

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Predictors	Model	Mean	Std	Min	Max
Eye	Random Forest	0.802	0.061	0.679	0.882
	XGBoost	0.833	0.051	0.741	0.898
	ROCKET	**0.916**	**0.039**	**0.842**	**0.977**
Vehicle	Random Forest	0.848	0.032	0.798	0.883
	XGBoost	0.865	0.039	0.788	0.902
	ROCKET	**0.880**	**0.034**	**0.818**	**0.925**
Combined	Random Forest	0.852	0.062	0.713	0.922
	XGBoost	0.902	0.038	0.830	0.952
	ROCKET	**0.955**	**0.027**	**0.906**	**0.986**

TABLE 3 Performance summaries by modeling approach and available predictors for 90-second intervals.

Data Type	Window (s)	1st	2nd	3rd	4th
Eye	15	skew_left_pupil_size 0.157 (0.045, 0.238)	kurt_left_pupil_size 0.051 (0.001, 0.256)	sd_head_pos_z 0.026 (0.004, 0.060)	middle_90_head_pos_y 0.026 (0.006, 0.059)
	30	skew_left_pupil_size 0.102 (0.050, 0.187)	kurt_left_pupil_size 0.069 (0.007, 0.156)	sd_left_pupil_size 0.032 (0.003, 0.159)	sd_head_pos_z 0.029 (0.004, 0.080)
	45	kurt_left_pupil_size 0.090 (0.004, 0.225)	skew_left_pupil_size 0.076 (0.033, 0.148)	middle_90_head_pos_y 0.051 (0.026, 0.099)	middle_90_head_pos_z 0.033 (0.012, 0.064)
	60	skew_left_pupil_size 0.075 (0.036, 0.120)	kurt_left_pupil_size 0.053 (0.003, 0.202)	kurt_right_pupil_size 0.029 (0.000, 0.161)	skew_right_pupil_size 0.028 (0.006, 0.073)
	75	skew_left_pupil_size 0.074 (0.031, 0.123)	kurt_left_pupil_size 0.044 (0.003, 0.136)	skew_right_pupil_size 0.031 (0.003, 0.113)	middle_90_head_pos_y 0.029 (0.008, 0.068)
	90	kurt_left_pupil_size 0.084 (0.001, 0.190)	skew_left_pupil_size 0.064 (0.030, 0.105)	sd_head_pos_y 0.045 (0.019, 0.101)	middle_90_head_pos_z 0.029 (0.003, 0.073)
Vehicle	15	sd_steer_angle_rate 0.109 (0.045, 0.172)	q95_speed 0.053 (0.022, 0.099)	q95_accel 0.050 (0.019, 0.101)	sd_steer_angle 0.048 (0.013, 0.114)
	30	sd_steer_angle_rate 0.193 (0.113, 0.261)	middle_90_lane_dev 0.047 (0.028, 0.061)	q95_speed 0.045 (0.024, 0.073)	sd_lane_dev 0.036 (0.016, 0.055)
	45	sd_steer_angle_rate 0.148 (0.091, 0.252)	sd_lane_dev 0.064 (0.036, 0.095)	q95_speed 0.054 (0.023, 0.107)	q5_steer_angle_rate 0.045 (0.013, 0.161)
	60	sd_steer_angle_rate 0.160 (0.100, 0.235)	sd_lane_dev 0.067 (0.022, 0.126)	sd_accel 0.050 (0.019, 0.102)	middle_90_lane_dev 0.047 (0.030, 0.075)
	75	sd_steer_angle_rate 0.155 (0.056, 0.301)	sd_accel 0.077 (0.019, 0.169)	sd_lane_dev 0.062 (0.035, 0.101)	middle_90_steer_ang 0.049 (0.010, 0.127)
	90	sd_steer_angle_rate 0.110 (0.053, 0.283)	sd_lane_dev 0.089 (0.028, 0.240)	sd_steer_angle 0.062 (0.008, 0.152)	middle_90_steer_ang 0.060 (0.012, 0.156)
Combined	15	skew_left_pupil_size 0.096 (0.022, 0.215)	sd_steer_angle_rate 0.062 (0.030, 0.129)	middle_90_steer_angle_rate 0.038 (0.001, 0.174)	sd_steer_angle 0.036 (0.000, 0.133)
	30	skew_left_pupil_size 0.086 (0.032, 0.173)	sd_steer_angle_rate 0.051 (0.024, 0.077)	kurt_left_pupil_size 0.030 (0.002, 0.121)	middle_90_steer_angle_rate 0.023 (0.001, 0.084)
	45	skew_left_pupil_size 0.093 (0.027, 0.197)	sd_steer_angle_rate 0.042 (0.021, 0.063)	kurt_left_pupil_size 0.026 (0.004, 0.045)	middle_90_steer_ang 0.024 (0.001, 0.206)
	60	skew_left_pupil_size 0.061 (0.018, 0.111)	sd_steer_angle_rate 0.049 (0.026, 0.095)	skew_right_pupil_size 0.029 (0.000, 0.173)	middle_90_steer_angle_rate 0.029 (0.002, 0.195)
	75	skew_left_pupil_size 0.060 (0.018, 0.123)	sd_steer_angle_rate 0.045 (0.021, 0.083)	sd_steer_angle 0.040 (0.001, 0.252)	sd_xgaze_left 0.030 (0.000, 0.140)
	90	skew_left_pupil_size 0.054 (0.014, 0.132)	middle_90_steer_ang 0.052 (0.000, 0.269)	sd_lane_dev 0.043 (0.009, 0.082)	q95_steer_angle 0.032 (0.000, 0.235)

TABLE 4 Average feature importance and their ranges(max, min) for the 4 most important features by predictor type and interval length for XGBoost models

Under Review